

Introductory Guide

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1 Introduction

This user manual serves as a comprehensive guide for operating the key instruments used in nanofabrication and characterization. It is intended to support new users, researchers, and students in understanding both the principles of operation and the step-by-step operation procedures of various advanced systems like lithography, etching, deposition.

The manual begins with detailed sections on Raith Electron Beam Lithography (EBL) and Plassys MG 200 Etching System, explaining their core working principles followed by systematic operating procedures. It also includes instructions for maskless photolithography using Heidelberg Instruments (μ MLA), and thin film deposition techniques with the Plassys system.

Subsequent sections introduce a range of essential laboratory equipment, including the wire bonder, Bruker Atomic Force Microscope (AFM), and Horiba Raman Spectrometer, followed by critical tools for electrical measurements such as Keithley, Lock-In Amplifier, and Vector Network Analyzer.

Finally, the manual covers various vacuum and cryogenic components—rotary and turbomolecular pumps, circulators, and passive components like bias tees, attenuators, to provide a complete reference.

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2 Raith Electron Beam Lithography (EBL) System

System Principle



Figure 1: Raith Electron Beam Lithography System

Electron Beam Lithography (EBL) is a high-precision nanofabrication technique that uses a focused beam of electrons to directly write intricate patterns onto a resist-coated substrate. Unlike traditional photolithography, which relies on optical masks to define patterns, EBL is a maskless, direct-write method that offers extremely high resolution—capable of producing features smaller than 100 nanometers. This makes it especially valuable in nanotechnology, research, and advanced prototyping applications where custom and ultra-fine structures are needed.

The EBL system consists of several key components. At the core is the electron gun, which generates a high-energy electron beam, typically in the range of 10–100 keV. This beam is focused and shaped by a system of electron optics, including electromagnetic lenses, which control beam diameter and intensity. A beam deflection system accurately steers the beam across the surface of the resist-coated substrate in precise patterns. The substrate is coated with an electron-sensitive resist, such as PMMA (polymethyl methacrylate), which responds chemically to the electron exposure. A high-precision stage holds and moves the substrate to align it with the beam during writing, while a control system (GDS file) synchronizes beam movement and pattern data for accurate exposure. In the exposure step, during which the finely focused electron beam is scanned over the resist surface based on a pre-designed digital pattern. The energy from the electrons causes localized chemical changes in the resist. In positive resists like PMMA, the

high-energy electrons break the polymer chains, making the exposed regions more soluble in a developer solution. In contrast, negative resists such as HSQ undergo cross-linking, where exposed regions become less soluble and remain after development.

Aspect	Positive Resist	Negative Resist
Exposure Effect	More soluble after exposure	Less soluble after exposure
Development Result	Exposed areas are removed	Unexposed areas are removed
Mask Pattern	Matches the mask	Inverse of the mask
Resolution	Higher resolution	Lower resolution
Application Preference	Used for fine patterns	Used for thicker layers
Exposure Reaction	Polymer chains break	Polymer chains cross-link
Ease of Use	Easier to control	More sensitive to variations

In electron beam lithography (EBL), the precise delivery of charge to the resist layer is critical for achieving accurate patterning. This charge delivery is governed by the beam current and the exposure time, linked by the relationship

$$D \times A = I \times \Delta T,$$

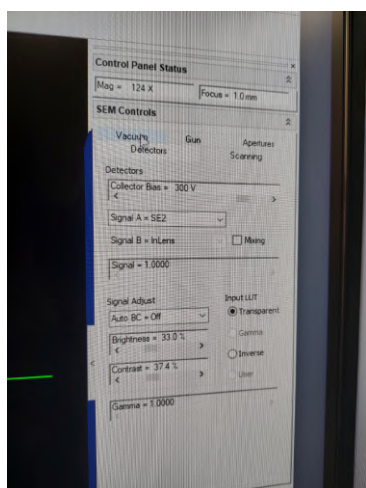
where D is the dose in $\mu\text{C}/\text{cm}^2$, A is the area of exposure, I is the beam current (typically measured in picoamperes), and ΔT is the exposure time. Under typical settings such as an accelerating voltage of 25 keV, a 30 μm aperture, and a 10 mm working distance—the beam current usually falls between 220 pA and 300 pA, and this value must be measured precisely using a Faraday cup. This current determines how long the beam should dwell on a given spot to deliver the desired dose. Importantly, the exposure does not occur over the entire pattern simultaneously; instead, the electron beam writes spot-by-spot. Therefore, the area and time in the dose equation refer to a *single exposure spot*, not the full pattern. The final exposed structure is built up by the accumulation of many such spots. The user is responsible for defining the spot size, and based on that and the measured beam current, the system calculates the required dwell time to ensure each spot receives the correct dose. Accurate control of these parameters ensures uniform exposure and high-resolution pattern fidelity.

Following exposure, the sample undergoes development, where it is immersed in a chemical solution (deionized water and IPA in a 3:1 ratio) that selectively dissolves either the exposed or unexposed areas, depending on whether a positive or negative resist was used. This reveals the patterned resist structure on the substrate.

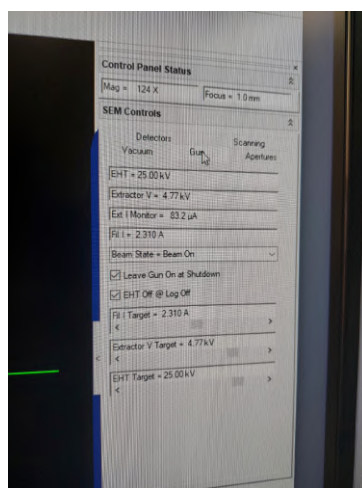
Operating Procedure

1. Place the sample on the substrate holder/tray. Put a few gold nanoparticles (up to 20nm in diameter) on the substrate to help in adjusting focus and stigmatism.
2. Open the door of the loading chamber and place the tray in the socket.
3. Close the door and turn the knob tightly.
4. Open the Eline software and follow the instructions for the load lock process.

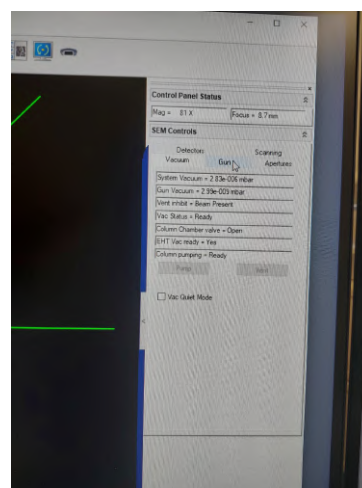
5. Open the load lock and pull it out. Gently insert the transfer rod. You will be able to see the substrate holder in the main chamber on the computer screen.
6. Remove the transfer rod gently. Close the load lock.
7. A window appears that asks: Reset Coordinate System. Click Yes.
8. On the right screen on the XY UV icon, select Base: XY, Length: mm, and set Z= 27mm, position = absolute.
9. On the left screen. Click on the Gun and set EHT = 25kV.
10. For Aperture set 30 μm standard
11. For Detector set Signal A = SE2 and Signal B= InLens.
12. Use Ctrl+Right click to go to any position on the stage. Go towards the edge on the substrate that has to be exposed.



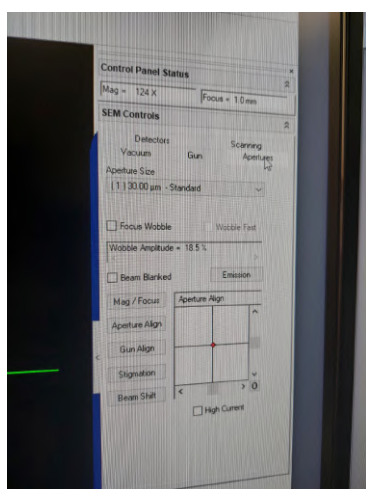
(a)



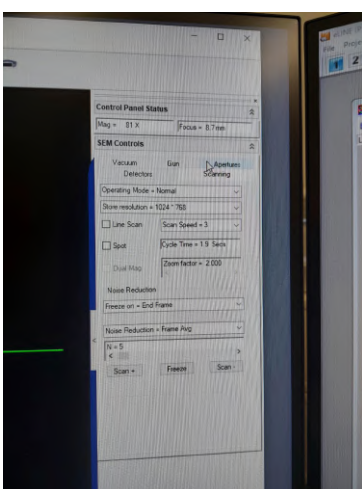
(b)



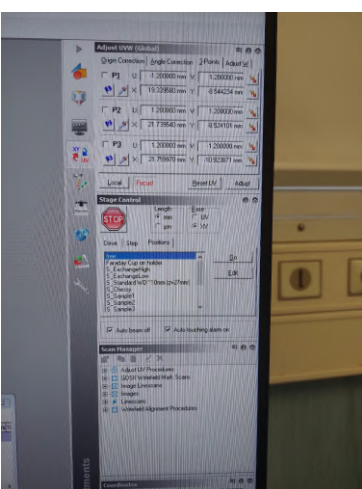
(c)



(d)



(e)

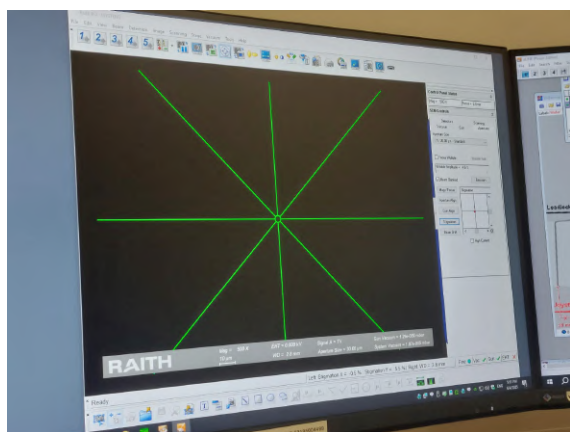


(f)

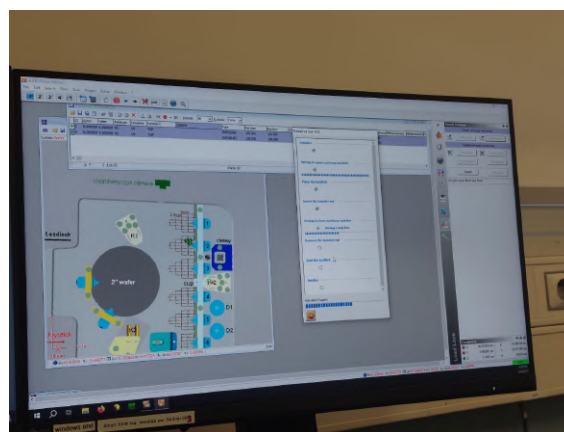
Figure 2: Settings for (a) Detector (b) Gun (c) Vacuum (d) Aperture (e) Scanning (f) Three point alignment

13. Change Coarse on left screen to Fine. Working Distance is 9.6mm. An inverted pentagon on the right screen is the icon to switch the beam on/off.
14. Use magnification and focus buttons to focus.
15. Play with contrast and brightness to improve the image.
16. In Aperture, click on stigmation align (X and Y both can be done separately) and try to focus on the gold nanoparticle that was placed close to the edge of the substrate.
17. Similarly, do for aperture. Tune both X and Y knobs until you see fine features of nanoparticles.
18. Use the icon "Reduced raster/apertures" to focus on small areas.
19. One can start focusing in 40 or 50kX magnification (100 nm).
20. In aperture, go to focus wobble: play with wobble amplitude to make improve features on the image. Wobble fast should be on this time. During this process, the motion of the image should be up and down and not in the X and Y direction.
21. One can start focusing at 150 or 160 kX magnification (20nm). The fine features of nanoparticles should be focused well at this stage.
22. During operation Gun vacuum = $2.87e^{-9}$ mbar and System vacuum = $2.76e^{-6}$ mbar.
23. In stage control, select "Faraday cup on holder" and press Go. Demagnify to see the cup.
24. Turn on the beam and magnify (magnification should be corresponding to 10nm). Go to the patterning parameter and measure the beam current. Note this value. The current should be around 0.24 or 0.25 nA.
25. In order to see the image, you can click on the icon "IMG" next to the beam on/off icon.
26. Find the arrow marker on the chip/substrate. The next step is to do a four-point alignment.
27. In Scan Manager, go to writefield alignment procedures (right screen). Select Manual and then select 200 μ m WF - 25 μ m marks (Manual ALWF). Then correct 4 points with the left click and execute. After all four points are done, select 200 μ m WF - 5 μ m marks (Manual ALWF). Center align all four points and similarly do for 200 μ m WF - 1 μ m marks (Manual ALWF).
28. There are four markers on the chip at each corner. Each marker has two squares. Move to the left uppermost marker. Don't change the focus at this stage, but you can play with magnification to help in finding the marker using the joystick.
29. Click on XY UV icon. On the Adjust UVW(Global) window, click on Adjust. In stage control, see if the step size is 2.4 mm for X and Y. For P1, click on the pin to set coordinates, and then tick mark.

30. Then click on the right arrow to move to the alignment marker on the upper right side. Similarly, click on the pin to load coordinates and then tick mark to set the values. Now, click on the bottom arrow to move to the alignment mark on the bottom right side of the substrate (the pointer should ideally move to the marker). Similarly, set the coordinates for the same and click on the tick. Click on the drive to destination to see that all three markers are in place. Lastly, click on Adjust. Note: One can click on IMG or pattern to see the marker or unsee it, respectively.
31. Click on the geometrical icon consisting of a triangle, a circle, and a square on the right screen and load the GDS file that contains the pattern that needs to be written.
32. On top left, click on edit. There will be a blue and white icon to select the working area palette (in the editor). Put the areas as 200, 200, 200. You can see coordinates and set them to zero.
33. Select GDS II layers to see the whole design and save it. Drag "top" on the position list. Add the layer that is to be scanned. Start with the default or marker layer number 63. Go to properties and set dose, step size. Click on the position every time to set 0,0 as the origin.
34. Click on the patterning parameter and then go to the calculator. Calculate everything. Check the layer by clicking go open to see if the correct layer will be exposed.
35. Press scan to expose the selected layer.
36. Turn off EHT (left screen). Place the beam on K2.
37. Finally, follow the instructions to unload the sample.



(a) Left screen window



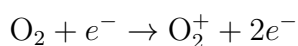
(b) Right screen window

3 Plassys MG 200 Etching System

System Principle

The Reactive Ion Etching (RIE) is a plasma-based dry etching technique used in micro-fabrication to etch precise patterns on a substrate. RIE works by creating a low-pressure plasma from a reactive gas mixture inside a vacuum chamber. This plasma contains ions and reactive radicals that physically bombard the surface (via ion momentum, like sputtering) and chemically react with materials to form volatile byproducts that are pumped away. The combination of both physical sputtering and chemical reactions enables high selectivity and directional etching. Plasma is generated through the following process:

1. A low-pressure gas (e.g., O₂, CHF₃, SF₆) is introduced into the chamber.
2. A radio frequency (RF) electric field (typically 13.56 MHz) is applied across the electrodes.
3. This RF field accelerates free electrons in the gas.
4. These high-speed electrons collide with gas molecules and ionize them. For example:



5. As a result, a plasma is formed — a glowing, reactive mixture consisting of:
 - Free electrons
 - Positive and negative ions
 - Neutral atoms or molecules
 - Excited species and radicals

Operating Procedure

1. Open valves for Compressed Air, Oxygen, Nitrogen, and Water.
2. Press the Isolement Button to create a vacuum inside the chamber (green light off).
3. Wait until the pressure inside the chamber goes to 7.8×10^{-5} . The Rotary Pump is activated first, and consequently, the Turbo Pump starts automatically.
4. Click on the Process button (green light on). Turn the oxygen knob and set flow rate to 10 sccm.
5. Open the butterfly valve to set a pressure of 0.025 mbar inside the chamber.
6. Turn on the AC power supply and then turn on the RF Power button.
7. Tune the knob to set a voltage corresponding to 144 V on the voltage gauge (see the monitor). Wait for 3 minutes.



(a)



(b)



(c)



(d)



(e)

Figure 4: (a) Butterfly Valve to set pressure inside the etching chamber (b) Nitrogen Gas Inlet (helps in pumping and purging) (c) Knobs to set flow rate for gas that will be used for etching (d) Pressure Monitor and Pumping/Isolation switches (e) Switch to set RF Power.

8. Turn off the RF button. Close the butterfly valve completely. Close the oxygen valve and set the flow rate to 0 sccm. Don't turn off the AC power supply. Click on the Process button (green light off). Wait for 2 minutes.
9. Now open the nitrogen inlet valve until the pressure on the gauge shows 0.5 mbar. Click on isolement (green light on). The pressure drops to 0 on the gauge. Again, open the valve and click on isolement (green light off). Do this two times. This is called pumping and purging. This is done in order to clean the chamber. For the third time, open the inlet valve completely, and the chamber door will be open.
10. Now you can place the sample inside the chamber and close the door.
11. Click on isolement (green light off). The door will get locked automatically due to the creation of a vacuum. Wait until pressure goes to 7.8×10^{-5} . Then click on process. Set the oxygen flow rate, set the chamber pressure.

12. Click on the RF button (green light on) for 30 seconds or 60 seconds until you want to do the etching. Press on RF button to stop etching (green light off).
13. Reverse the procedure to get the samples out of the chamber. Pump the chamber before leaving for the next user.

Substance	Role in RIE System	Description
Compressed Air	Mechanical/pneumatic control	To actuate valves or move mechanical parts. Not used in the etching.
O ₂	Etching + cleaning	To etch organic materials (like graphite) and for plasma cleaning.
N ₂	Pumping and Purging/venting/dilution	Evacuating(pumping out) the chamber and then refilling it (purging) with an inert gas (N ₂ or Ar) to remove residual gases or contaminants.
H ₂ O	Cooling (indirect)	Used in chiller units to control electrode or chamber temperature. Helps in cooling electronics. Not directly introduced into the chamber.
CHF ₃	Etching of SiO ₂ / polymers	Provides fluorine radicals for SiO ₂ and hBN etching. Often combined with O ₂ .
SF ₆	Etching of silicon	Used for Si etching. Produces reactive fluorine species.

Table 1: Roles of Common Gases and Fluids in Reactive Ion Etching (RIE)

4 Heidelberg Instruments Maskless Optical Photolithography (μ MLA)

System Principle

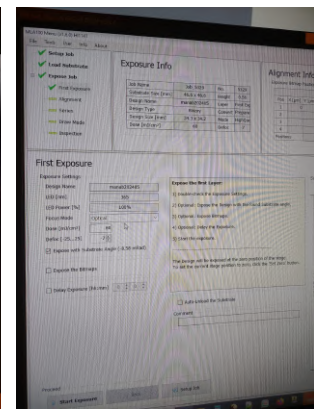
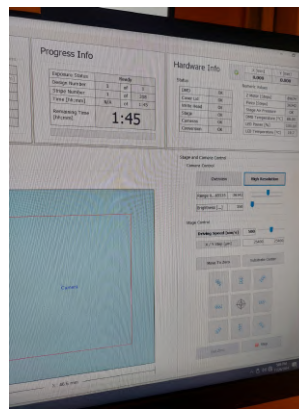
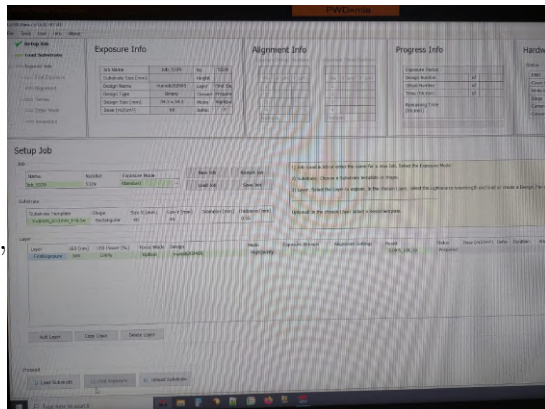
μ MLA operates on the principle of direct-write photolithography using digitally controlled light projection, eliminating the need for traditional photomasks. Instead of using fixed photomasks to define patterns on a photoresist-coated substrate, the μ MLA system employs a Digital Micromirror Device (DMD) an array of tiny tiltable mirrors that can individually reflect light in precise directions. Each mirror represents a pixel in the exposure pattern, and by selectively tilting these mirrors, the system projects a computer-defined pattern directly onto the substrate. A UV light source, typically operating at wavelengths such as 375 nm or 405 nm, is used to expose the photoresist. The optical system focuses and projects the modulated light pattern onto the substrate with high resolution and uniformity.

The substrate itself is mounted on a high-precision motorized stage that moves in the X and Y directions, allowing for stitching of large patterns that exceed the single field of view of the DMD. The process begins by preparing the substrate with a uniform coating of photoresist, followed by uploading a digital design (GDSII or DXF file) into the control software. The system then aligns the substrate, adjusts focus, and initiates exposure by synchronizing the DMD with the stage movement. Depending on the resist type, exposed regions are either removed (positive resist) or retained (negative resist) during the development process. This technique allows for rapid prototyping, maskless operation, and multi-layer patterning with high precision.



Figure 5: μ MLA

Operating Procedure



(a) Window Before Starting Procedure (b) During Exposure (c) Steps involved

1. Setting up Exposure Job: Click on New Job or Load Job to load from a list of previously saved jobs. The name of the job can be changed and saved. The number of the job is fixed.
2. Select Standard Exposure (without overlay) to expose a single design for each layer. In order to expose the layer again one can click on repeat job.
3. In the substrate frame: Double click on the substrate and load the substrate. Size X(mm): 38mm, Size Y(mm): 38mm, Shape: Rectangular, Thickness: 0.525mm
4. In the Layer frame: Double click design and load the GDS design file. Select the Resist type: *S1805_LOL2000_SiO₂-Si*, Defoc: 7, Dose: 50.
5. Click on the load Substrate button on the bottom left. The system will start the search for the center of the substrate.
6. Click on manually find the center and press on the icon that has two concentric circles and a cross wire.
7. At this stage click anywhere on the screen and press the accept position. A window will appear, and then press continue.
8. Find the plate center window will appear with X and Y measured.
9. Click on overview to check focus and then check resolution. Check the focus of the spot. Click on Tools and press on Large Camera defocal. Change the "Change range" to focus more on the dot.
10. In the first exposure panel double-check the design name, dose, and defoc.
11. After verifying everything start exposure.

5 Plassys MEB550S Evaporator

System Principle

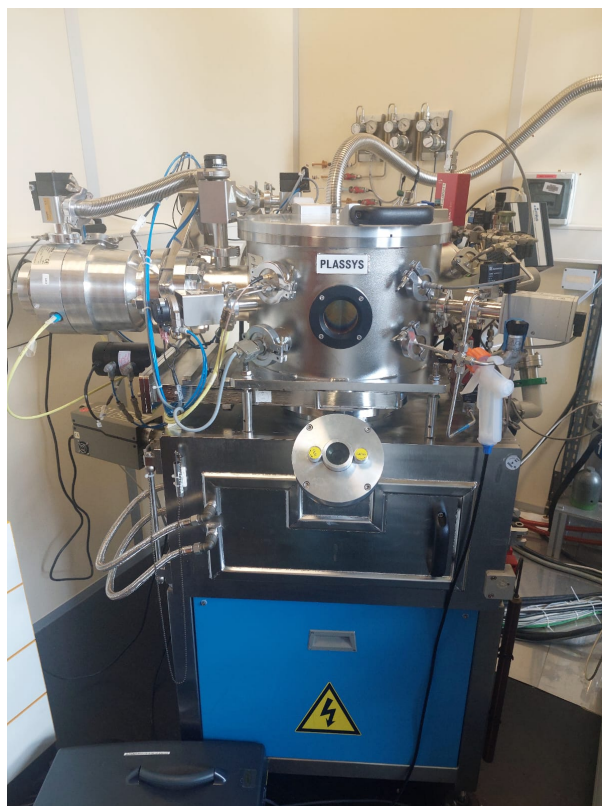


Figure 6: Plassys Deposition System

The **Plassys MEB550S Evaporator** is a high-vacuum deposition system used for thin-film metal deposition in micro- and nanofabrication processes. Its operation is based on the *physical vapor deposition (PVD)* principle, specifically utilizing **electron-beam evaporation** to deposit materials such as chromium, titanium, gold, copper, and palladium with high precision and purity. This process takes place in a controlled vacuum environment, which is critical to ensure the clean transfer of material from the source to the substrate without contamination or oxidation.

At the heart of the system lies the **evaporation source system**, which houses the electron beam (e-gun) that bombards individual metal sources held in separate crucibles within the loadlock chamber. When the high-energy electron beam is directed onto a target metal, it vaporizes the material, and the resulting metal vapor travels upwards to condense onto the substrate located inside the evaporation chamber. Each crucible holds a different metal, allowing the user to perform single or multi-layer deposition in one vacuum cycle without breaking the vacuum. The e-gun produces intense localized heating, allowing even high-melting-point metals to be evaporated efficiently.

The **vacuum system** of the Plassys MEB550S is critical to its operation. It maintains an ultra-high vacuum (typically below 10^{-7} mbar) in both the loadlock and evaporation chambers. A combination of *rotary pumps*, *turbo-molecular pumps*, and *loadlock isolation*

valves is used to reach and sustain these low pressures. This vacuum environment is necessary to minimize collisions between evaporated atoms and residual gas molecules, thereby ensuring uniform and uncontaminated thin film deposition.

A **Quartz Crystal Microbalance (QCM)** is integrated into the system to monitor the deposition rate and film thickness in real time. The QCM measures mass accumulation by tracking changes in the resonant frequency of a quartz crystal. This feedback allows precise control of deposition parameters and is vital for achieving consistent and reproducible thin films.

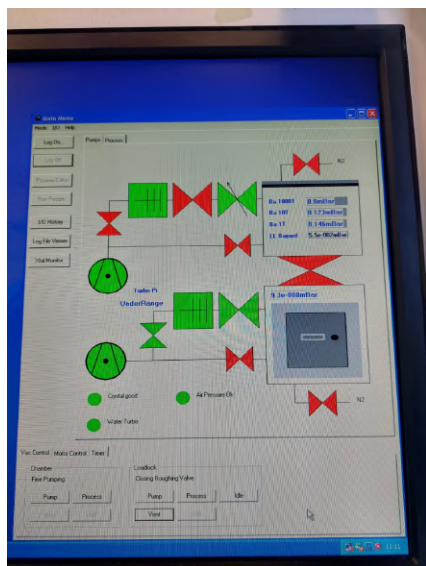
The **control panel**, located in the front section of the system, enables users to operate and monitor the entire process. The *computer interface* provides access to real-time pressure readings, deposition thickness, rate control, and execution of custom evaporation recipes. Users can also open or close the evaporation chamber, toggle the e-gun, and manage all essential parameters through this interface.

Within the **evaporation chamber**, a *rotating substrate fixture* is installed. This fixture allows the substrate holder to rotate and tilt during deposition, enabling control over the deposition angle and improving film uniformity. The substrate holder itself, shown in Figure 1.11D, includes clips or wire mounts that secure the substrate and prevent it from falling into the chamber during the process. This holder is mounted on the rotating fixture (as seen in Figure 1.11E), ensuring alignment and stability throughout the operation.

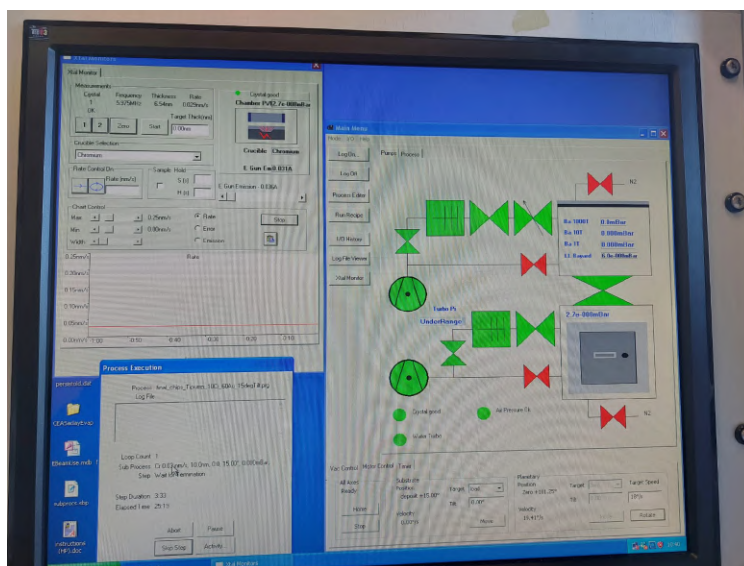
An important feature of the system is the **loadlock valve**, which separates the loadlock and evaporation chambers. It typically remains closed during the ramp-up phase and only opens once the desired deposition rate has been achieved. In certain cases, such as *titanium deposition*, the valve is deliberately kept closed to allow titanium vapor to react with residual oxygen in the loadlock chamber. This reaction helps to further reduce the background pressure, improving the vacuum quality before actual deposition begins. Evaporating a small amount of titanium before actual deposition serves as a getter process to remove residual reactive gases like oxygen and water vapor from the vacuum chamber. Titanium, being highly reactive, binds with these gases, helping to clean the chamber and improve vacuum quality—especially important when achieving ultra-high vacuum is difficult. Reducing chamber pressure is critical in electron-beam evaporation to minimize scattering of evaporated atoms, enhance film purity, and ensure better adhesion and uniformity of the deposited film. Although the chamber pressure is generally uniform, titanium gettering is typically performed in the loadlock chamber before it's connected to the main evaporation chamber, preventing contamination and maintaining clean deposition conditions.

During operation, the e-gun's bombardment of the metal source produces a **distinctively bright and colored light** (illustrated in Figure 1.11F), which is visually indicative of the evaporation process. The color and intensity vary depending on the metal being evaporated. This visual feedback, combined with the monitored parameters, helps the user ensure that the system is functioning correctly. In summary, the **Plassys MEB550S** combines electron-beam precision, advanced vacuum technology, and real-time process control to provide a reliable platform for high-quality thin film deposition.

Operating Procedure



(a) Pumping process



(b) Deposition step

1. Place the sample on the sample holder and check if it's firmly placed by inverting the sample holder. Place the sample holder in the chamber.
2. Pump the chamber overnight until the pressure drops to 10^{-7} mbar. The electron gun should be active during this time, which is ideally the case.
3. Before evaporating the required metal for our sample, the pressure in the chamber can be decreased and stabilized by evaporating titanium while keeping the valve between the loadlock and metal crucibles closed in order to avoid deposition on the final substrate.
4. Titanium can be evaporated at a controlled rate of 0.2nm/s up to a maximum thickness of 100nm. This process effectively reduces the chamber pressure by a factor of 2 to 4, due to the strong gettering action of titanium, which binds with residual gas molecules, particularly oxygen.
5. Start the process by loading the evaporation recipe.

6 Wire Bonder

Wire bonding is a method used to establish electrical interconnections between an integrated circuit (IC) chip and its external leads or substrate using extremely fine metallic wires, typically composed of gold (Au), aluminum (Al), or copper (Cu). The bonding process begins by placing the semiconductor chip on a lead frame or a substrate, where it is placed on a stage whose temperature is set to 135 degrees. Once secured, wire bonding can commence using one of the two primary techniques: thermosonic ball bonding or wedge bonding. Ball bonding, which is widely used with gold wires, is the focus of this explanation.

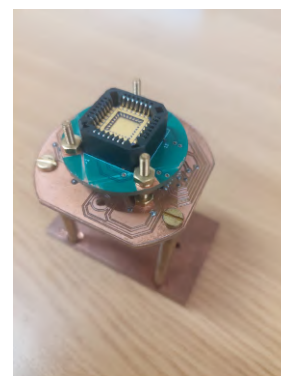
In the ball bonding system, the bonding wire is passed through a capillary, and the end of the wire is melted using a No Electronic Flame-Off (NEFO) process to form a Free-Air Ball (FAB). This air-free gold ball becomes the bonding element. The first bond is created by precisely positioning the capillary over a designated bond pad on the chip or substrate. With carefully controlled pressure, ultrasonic vibration, and elevated temperature, the capillary presses the ball onto the surface. The ultrasonic energy serves to break through surface oxides or contaminants and initiate atomic diffusion, forming a metallurgical bond. Following the first bond, the capillary is elevated and moved to a second bonding location, typically on the package's inner lead or substrate pad. As the wire is continuously fed during this movement, a loop or arc is formed. Upon reaching the second location, the capillary descends to make the second bond—commonly referred to as a stitch or wedge bond—by again applying ultrasonic energy and pressure, forming a secure electrical and mechanical connection. The wire is then clamped and broken off to complete the cycle, and a new FAB is generated for the next bonding sequence.



(a)



(b)



(c)

Figure 8: (a) Manual Wire Bonding Machine (b) Practice session for ball bonding (c) Chip Carrier placed on Chip Holder after wire bonding. PMMA and epoxy.

The ball bonding system includes multiple control mechanisms to ensure consistency and precision. For instance, manual spark is used to initiate the spark for forming the ball by activating an electrode underneath the capillary. To avoid electrical discharges that could damage the device, the NEFO is switched off before the bonding cycle begins. Additionally, a control panel allows users to define the bonding parameters such as ultrasonic

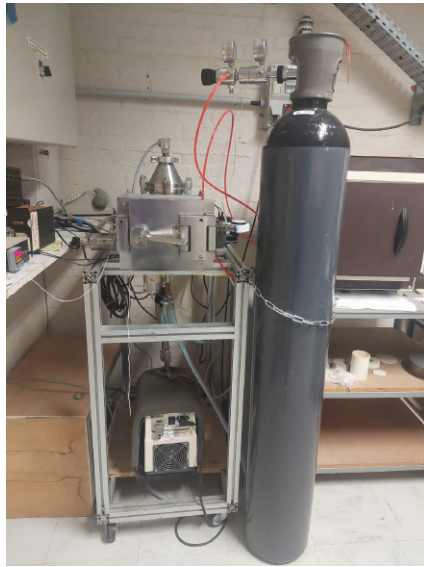
power, bonding force, and bonding time, which are critical for the quality of the bond.

A critical aspect of this system is the bonding stage, which is often composed of seven steps, starting from wire feeding and FAB formation, to first and second bond creation, and finally wire clamping and cutoff. Throughout the process, the chip carrier is often heated to a predefined temperature to aid in the bonding process. This thermal assistance, combined with ultrasonic energy, helps establish strong and reliable electrical and mechanical interconnections. In contrast, wedge bonding employs a different bonding tool shape and does not involve FAB formation. Instead, the wire is flattened and pressed onto the bond pads, relying on pressure and ultrasonic energy—usually at lower temperatures—which is especially beneficial for temperature-sensitive devices or fine-pitch applications. The differences between them are tabulated in the table below:

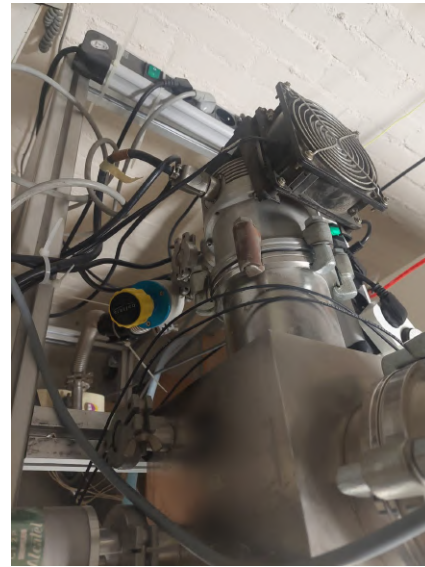
Feature	Ball Bonding	Wedge Bonding
Wire Material	Typically gold or copper	Usually aluminum (Al) or gold
Bonding Tool	Capillary tool	Wedge-shaped tool
Bond Shape	First bond is a ball; second is a wedge	Both ends are wedge-shaped bonds (no ball)
Bonding Mechanism	Combination of heat, pressure, and ultrasonic energy (thermosonic bonding)	Ultrasonic wedge bonding (cold weld); sometimes thermal compression is used
Typical Use	High-volume production, especially for gold wire	Low-temperature or sensitive applications

Table 2: Comparison of Ball Bonding and Wedge Bonding

7 Annealing System



(a)



(b)



(c)



(d)

Figure 9: (a) Annealing System (b) Yellow Isolation Valve (c) Pressure Gauge and CFF 450 controller (d) Temperature (Program) Controller

1. Close the yellow isolation wall. Check if the pumps are turned off.
2. Keep the door handle annealing chamber in the open position.
3. Open the nitrogen inlet valve and cylinder valve until the door of the annealing chamber opens. Once the door is open, close both nitrogen valves.
4. Keep the samples on the substrate holder in the center.
5. Check that the thermocouple is in contact with the substrate holder. Close the door of the annealing chamber.

6. Open the yellow isolation valve and turn on rotary(backing pump). Check if the door is properly shut.
7. Once the pressure is below 10^{-1} mbar, turn on the turbo pump.
8. On the CFF 450 controller, press the start button to the left and the controller button to the right. Pump until the pressure drop to 10^{-5} .
9. On the temperature controller, set the program using PGM button. The arrow pointing in downward direction is the okay button, while the arrow pointing in the upward direction is the start button or the program.
10. The samples are getting annealed now.
11. Check if the temperature is below 50 degrees. Turn off turbo and close the yellow isolation valve. Turn off backing pump.
12. Once the turbo has slowed down vent the annealing chamber by opening both nitrogen valves.
13. Remove the sample and close the door. Nitrogen valve should be closed before taking off the sample.
14. Open yellow isolation valve and turn on backing pump. Once the pressure goes below 10^{-1} mbar close the yellow valve and turn off the backing pump.
15. Check that all valves and doors are closed.

8 Bruker Atomic Force Microscope (AFM)

System Principle

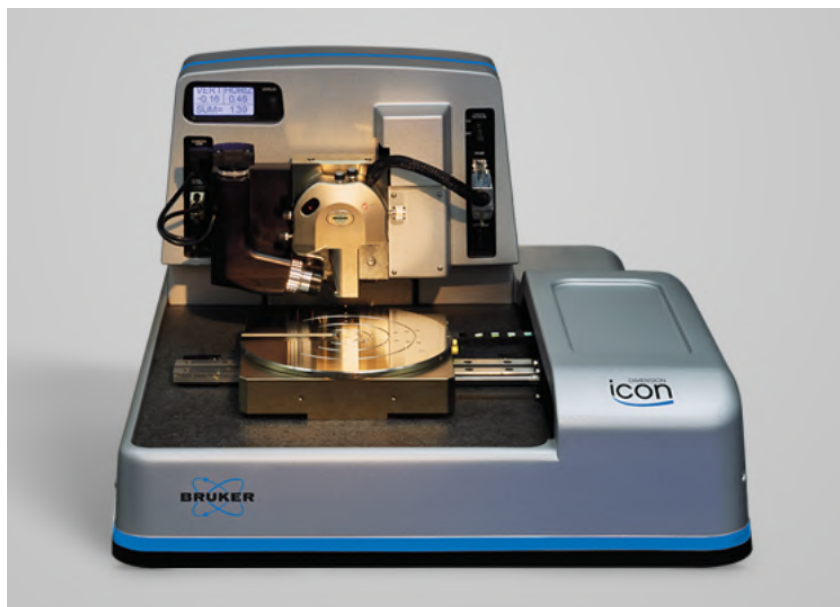


Figure 10: Bruker Atomic Force Microscope

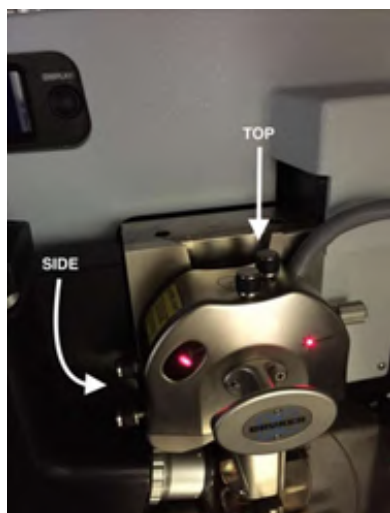
The Bruker Atomic Force Microscope (AFM) operates on the fundamental principle of measuring the interaction forces between a sharp probe (tip) and the surface of a sample at the nanoscale. The AFM uses a microscale cantilever with a nanometer-scale sharp tip, which is brought into close proximity with the sample surface. As the tip scans across the surface, various forces—such as van der Waals forces, electrostatic forces, magnetic interactions, or mechanical contact—act between the tip and the sample. These forces cause minute deflections of the cantilever, which are measured with high precision.

To detect the cantilever's deflection, the system uses an optical beam deflection method: a laser beam is focused onto the back of the cantilever and reflected onto a position-sensitive photodetector (usually a four-quadrant photodiode). As the cantilever bends due to interactions with surface features, the angle of the reflected laser beam shifts, and the resulting change in the photodiode signal is used to quantify the cantilever's deflection. This deflection corresponds to the topographical height variations on the sample surface.

The AFM can operate in several modes depending on the interaction mechanism and sample requirements. In Contact Mode, the tip physically touches the surface, and the deflection directly maps surface topography. In Tapping Mode (also called Intermittent Contact Mode), the cantilever oscillates near its resonant frequency, and the tip periodically touches the surface. The damping of the oscillation amplitude gives height information while reducing lateral forces, making it suitable for delicate samples. In Non-Contact Mode, the tip hovers just above the surface, sensing attractive forces without making contact, ideal for soft or sticky surfaces.

Bruker AFMs integrate advanced feedback control systems to maintain a constant force or oscillation amplitude between the tip and the surface, using a piezoelectric scanner to move the tip or sample with sub-nanometer precision in the x, y, and z directions. The height adjustments needed to maintain constant interaction force are recorded and used to construct high-resolution 3D surface maps.

Furthermore, Bruker AFMs often include advanced features such as phase imaging, force spectroscopy, and electrical/magnetic property mapping, making them powerful tools not only for imaging but also for probing material properties at the nanoscale.



(a) Scan Head



(b) Tip holder

Operating Procedure

1. Open Nanoscope software.
2. Select Tapping Mode and click on the green button to load the experiment.
3. In setup, select probe OTESPA R3. Click on save changes and return.
4. Move to the alignment station. Tune the knobs on the top of laser alignment station to maximize the voltage (4 - 4.23V). Using the knobs on the left side to bring the laser spot in the center with 0.0 or 0.1 V deflection in both horizontal and vertical directions.
5. Return from alignment station and focus on the tip until there is a good image.
6. Click on autotune to configure cantilever drive frequency and amplitude.
7. Go to Navigate and click on sample load position.
8. Load the substrate and turn on the vacuum. Locate the substrate exactly below the tip and see if the light is falling on the substrate by manually navigating to the region of interest.
9. Click on tip reflection. Zoom out and adjust illumination.

10. Lower the scan head until the sample is clearly focused. Now you can click on the sample (default).
11. Click on check parameters and adjust the proportional gain (3-5) and integral gain (1-2).
12. Click on Engage. The scanning will begin right now.
13. Go to scan and slowly increase the scan size from 0.0 μm to 50 μm at maximum.
14. Select folder and directory to save the scan.
15. Click on offset if there is a need to change the center and enter execute.
16. In order to use the AFM tip for cutting the sample, click on Nanoman. Click on path to draw the lines where one wants to cut. Then click on move to path and enter the Drive2 Amplitude value. Then press Do it.
17. In order to stop the scan, click on pause once the scan is complete and then withdraw the scan head multiple times until the Z coordinate changes from -17k to -12k at least.

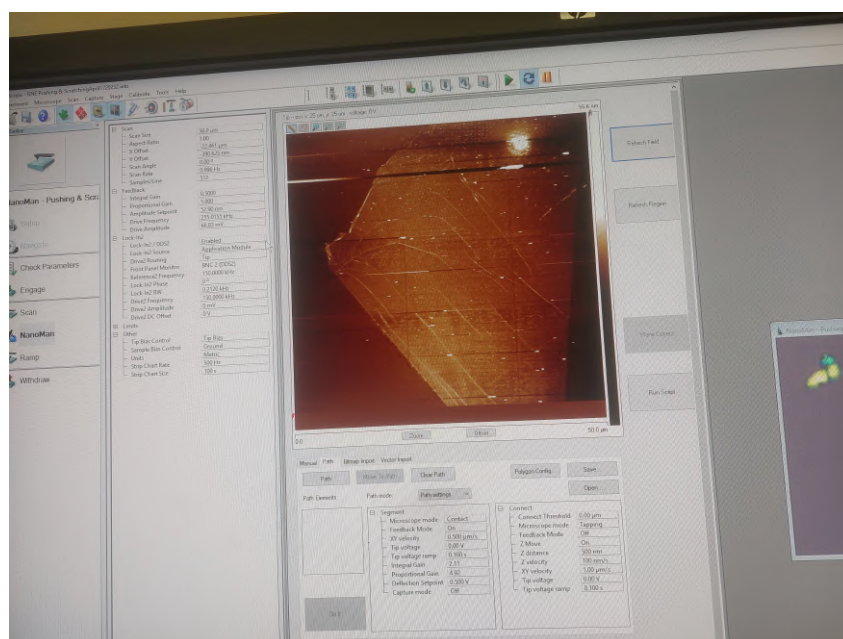


Figure 11: Cutting recipe for graphite/ multilayer graphene

9 Horiba Raman Spectrometer

System Principle

The Raman spectrometer operates on the fundamental principle of the **Raman effect**, which is a type of inelastic scattering of light that occurs when monochromatic light interacts with the vibrational or rotational modes of molecules within a sample. When a laser beam (typically in the visible or near-infrared range) is focused onto a sample, most of the incident photons are elastically scattered; this is known as Rayleigh scattering and involves no change in the photon's energy. However, a very small fraction of the photons—approximately 1 in 10^6 to 10^8 —are scattered inelastically, meaning their energy is shifted due to an exchange with the vibrational energy of the molecules. This shift gives rise to the Raman effect, where two forms of inelastic scattering can be observed: *Stokes scattering*, where the scattered photons have less energy because energy is transferred to the molecule, and *anti-Stokes scattering*, where the scattered photons gain energy by removing vibrational energy from the molecule.

The energy shift between the incident and scattered light is called the **Raman shift** and is typically expressed in wavenumbers (cm^{-1}). This shift corresponds to specific vibrational modes of the molecules in the sample, thereby providing a spectral fingerprint unique to the material. The Raman shift can be calculated using the formula:

$$\Delta\nu = \left(\frac{1}{\lambda_{\text{incident}}} - \frac{1}{\lambda_{\text{scattered}}} \right) \times 10^7$$

where the wavelengths are in nanometers. A typical Raman spectrometer comprises several key components that work together to produce the Raman spectrum. First, a stable monochromatic laser source is used to excite the sample. The laser light is directed onto the sample using focusing optics, and the scattered light from the sample is collected through a set of mirrors and lenses. Since Rayleigh scattered light is much stronger than the Raman scattered light, optical filters such as notch or edge filters are used to block the Rayleigh component and allow only the Raman-shifted light to pass through. This filtered light is then dispersed by a diffraction grating inside a spectrograph, which separates the light based on its wavelength. The dispersed light is finally detected using a sensitive charge-coupled device (CCD), and the data are processed by a computer to generate a Raman spectrum—typically a graph of intensity versus Raman shift.

This Raman spectrum reveals peaks that correspond to different molecular vibrations, with the position and intensity of these peaks offering insights into the chemical composition, molecular structure, and physical state of the material. The method is particularly powerful because it is non-destructive, requires minimal or no sample preparation, and can be used on solids, liquids, and gases. Moreover, since Raman spectroscopy detects changes in polarizability rather than dipole moment (as in IR spectroscopy), it is especially well-suited for analyzing non-polar bonds and aqueous solutions. Raman spectroscopy finds extensive applications in material science, chemistry, biology, pharmaceuticals, and even art conservation, providing a versatile and insightful technique for molecular characterization.

Operating Procedure

1. Switch on the power supply and turn on the laser source. Turn on the computer.
2. Open the Raman software "Labspec". Turn on the light source.
3. Select the grating 1200/600.
4. In the detector, set the CCD temperature -60 degrees.
5. Set the spectral centre to be 2700 cm^{-1} .
6. Set the laser power to be between 4 to 5mW.
7. Turn off the laser shutter and open the camera to find the sample placed on the sample stage.
8. Focus on 5x objective and then consequently move to 10x, 50x and 100x objective. Use the joystick to locate the sample on the substrate.
9. Turn off the camera and turn on the laser.
10. Set the acquisition time and start the measurement.
11. Use filtration and baseline correction to treat the data.
12. Save the data in ngs and text file format.

10 Probe Station

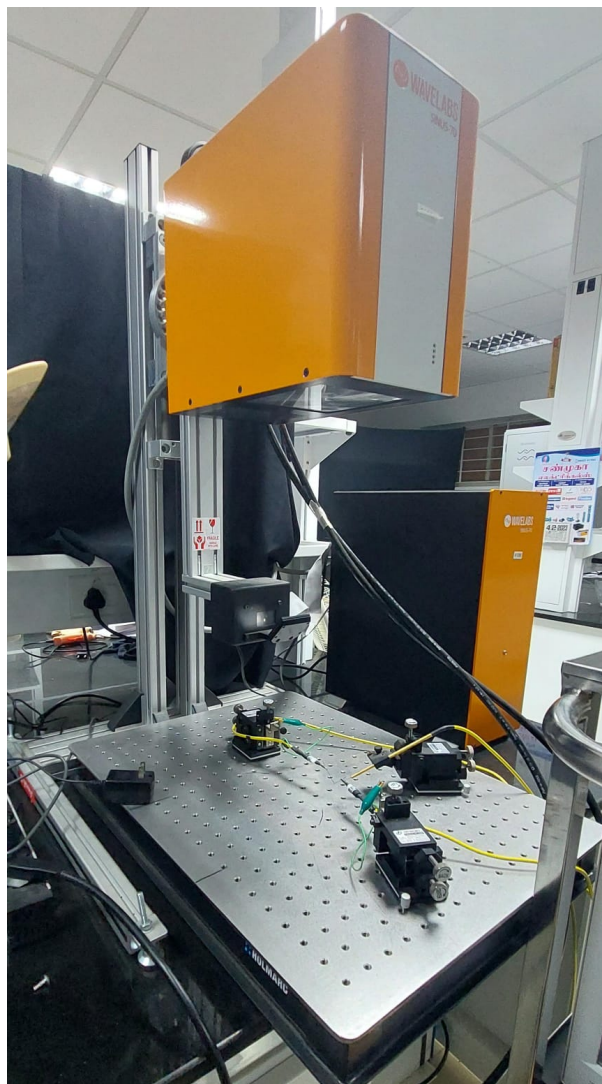


Figure 12: Probe Station

A probe station is a precision instrument used in semiconductor and electronic device research for testing and characterizing micro- and nano-scale devices fabricated on a substrate or wafer. It provides a controlled platform where electrical probes—typically fine-tipped metallic needles can be positioned with high accuracy to make temporary contact with metal pads or terminals on a device without the need for permanent wire bonding. Probe stations are widely employed in academic research labs, cleanrooms, and semiconductor industries to analyze device performance, evaluate fabrication quality, or debug circuits before final packaging.

The main purpose of a probe station is to establish reliable electrical connections to the internal circuitry of a microelectronic device through its contact pads so that various electrical characteristics can be measured using external instruments before loading the sample into the dilution refrigerator. These electrical measurements may include current-voltage (I-V) characteristics, resistance, capacitance, transistor operation, noise

analysis, and thermal response. In addition to development and process optimization, probe stations are also crucial in failure analysis and prototyping.

At its core, a probe station consists of a sample stage—often equipped with a vacuum chuck to hold the wafer or chip—alongside multiple micromanipulators that carry the probing needles. These micropositioners offer movement in the X, Y, and Z directions with micrometer-level precision. An optical or digital microscope can be integrated into the system to allow for accurate alignment of the probes with the sample's metal contact pads. Some advanced probe stations also include electromagnetic shielding, vibration isolation, or environmental chambers for more demanding testing environments.

The operation of a probe station begins by carefully placing the sample on the chuck, after which the user locates the contact pads under the microscope. Each probe is then maneuvered using the micropositioners until it makes gentle contact with the desired terminal. Once the electrical contact is established, external test equipment such as a source meter, voltage/current source, or semiconductor parameter analyzer can be connected to perform electrical characterization. If needed, the chuck can also be heated or cooled to examine the sample's behavior under varying thermal conditions.

To verify whether the sample is working properly, the probe station is used in conjunction with a DC source. The sample is first mounted on the chuck and aligned so that all relevant contact pads are visible and accessible. The probes are then carefully brought into contact with the sample pads using the micropositioners. Once a good contact is confirmed, the probes are connected to a DC source, such as a Keithley SourceMeter. Proper grounding and polarity must be ensured depending on the device architecture. Voltage or current sweeps are then applied across different terminals, and the response is recorded. The I-V characteristics are analyzed to determine the sample's behavior. For instance, a diode will exhibit rectifying behavior, while a resistor will display a linear response. A transistor will show gate-controlled modulation of current.

Before performing precise measurements, it is common practice to use a multimeter to check for continuity or resistance across the pads, confirming that the probes are properly connected and that there are no broken lines or shorts. Any deviation from the expected response could indicate fabrication defects, poor contact, or device failure.

In sensitive measurements, additional elements such as low-pass filters can be connected in-line to reduce high-frequency noise. Probe contact must be soft enough to avoid damaging the thin metal layers but sufficient to ensure good electrical conduction. Some probe stations also offer cryogenic capabilities for low-temperature testing or can be placed inside Faraday cages to shield against electromagnetic interference. The results obtained from probing are often used to generate I-V plots or other electrical performance graphs that are compared against theoretical or simulated models to evaluate the device.

11 Breakout Box

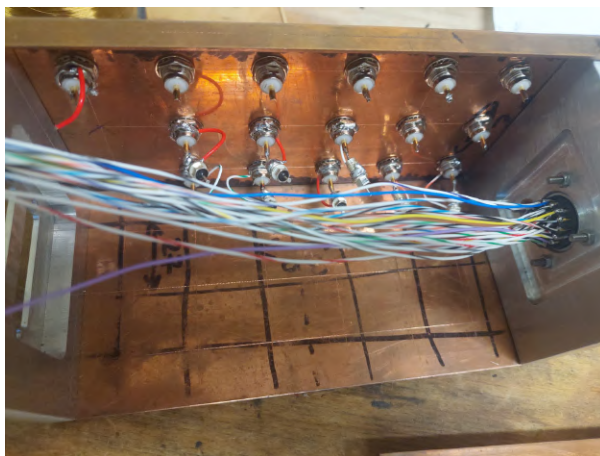


Figure 13: I learnt how a breakout box is made when a senior was soldering all the components. This is a picture of the same.

A **breakout box** is an interface tool used to simplify access to individual electrical connections within a multi-pin connector system, such as a Fischer connector. It breaks out each pin from a complex connector into discrete, accessible terminals—often with BNC connectors for easier monitoring measurements, or modification of each line in a circuit. Breakout boxes are especially useful in research setups and instrumentation, where multiple signal lines need to be routed, filtered, or debugged independently.

Connection to Fischer Connectors

Fischer connectors are circular, high-density, multi-pin connectors commonly used in scientific instrumentation due to their compact size, robust build, and secure locking mechanisms. They are used in cryogenic experiments as a part of the measurement electronics. When a breakout box is connected to a Fischer connector, it involves creating a custom adapter or interface cable that maps each pin of the Fischer connector to individual terminals on the breakout box. This mapping is done according to the pin configuration (often provided in a Fischer datasheet), ensuring each pin's signal is routed correctly to its corresponding output on the breakout box.

Inside the breakout box, the Fischer connector's cable typically terminates on a printed circuit board (PCB) or a set of terminal blocks. From there, each signal line can be routed through optional filtering components or left as a direct connection, depending on the design and purpose of the breakout box.

Types of Filters Used in a Breakout Box

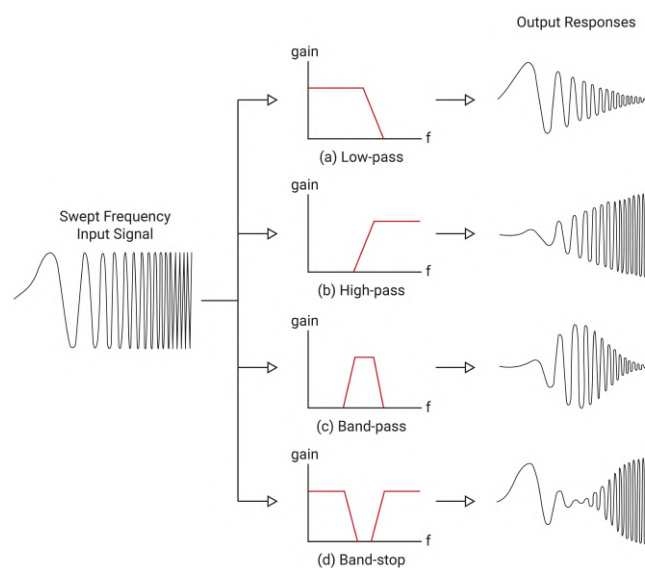
Breakout boxes are often enhanced with filters to condition signals by attenuating noise or separating frequency components. The types of filters that can be incorporated depend on the intended application, such as noise reduction, signal isolation, or frequency domain testing. Common types include:

RC (Resistor-Capacitor) Filters

These are the simplest and most commonly used analog filters. The cutoff frequency is determined by the formula:

$$f_c = \frac{1}{2\pi RC}$$

where R is the resistance and C is the capacitance. These filters are particularly useful for damping high-frequency pickup in sensitive measurement lines. The term RC filter is generic. Depending on how you connect R and C , it becomes a low-pass or high-pass filter.



(a) Frequency Response

Low-Pass Filters

Low-pass filters allow signals below a certain frequency to pass while attenuating those above. They are used when measuring DC or slowly varying analog signals, such as in temperature sensors or voltage measurements in low-noise environments.

High-Pass Filters

High-pass filters block low-frequency signals and pass higher frequencies. These can be useful when removing slow drifts or DC offsets from an AC signal, such as when analyzing high-frequency components in audio or RF systems.

Band-Pass Filters

Band-pass filters allow a specific range of frequencies to pass and attenuate both low and high frequencies outside this range. They are used in applications like spectroscopy or communication systems, where only a certain frequency band contains useful information.

Band-Stop (Notch) Filters

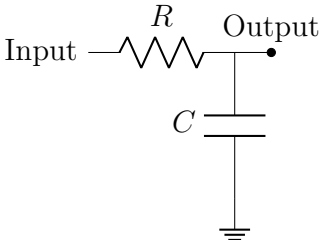
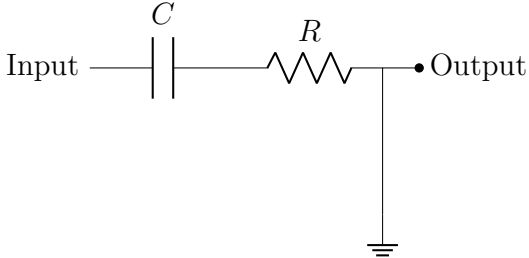
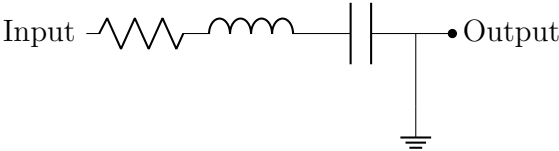
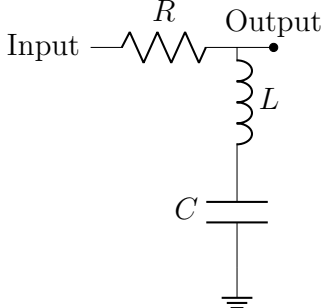
Also known as a band reject or notch filter, it blocks a specific range of frequencies while passing others. Notch filters may be included to remove specific interference frequencies

such as 50/60 Hz power line noise.

When designing or using a breakout box, attention must be paid to signal integrity, grounding, shielding, and cross-talk between lines. Breakout boxes with filtering are used in sensitive cryogenic experiments or high-precision electronic measurements. For example, in a dilution refrigerator setup, a breakout box might route signals from a Fischer connector to measurement instruments while applying low-pass filtering to suppress thermal noise or electromagnetic interference (EMI) picked up along long wiring paths. A typical passive filter, such as an RC (Resistor-Capacitor) filter, consists of two main terminals:

- **Input terminal:** Where the signal enters the filter.
- **Output terminal:** Where the filtered signal exits.

In addition to these two, most filter circuits include a ground connection. So in practice, a filter involves three nodes: input, output, and ground. The ground is necessary to provide a return path and to allow reactive components (like capacitors) to divert undesired frequency components. Filters are grounded through the breakout box's ground terminal. The breakout box provides a common ground shared by all connected instruments and filters. This grounding is essential to allow the capacitor in the RC filter to shunt high-frequency signals to ground and to maintain signal integrity and prevent noise or ground loops.

Low-Pass Filter (RC)	High-Pass Filter (RC)
	
Band-Pass Filter (Series RLC)	Band-Stop Filter (Parallel LC)
	

12 Rotary Pump

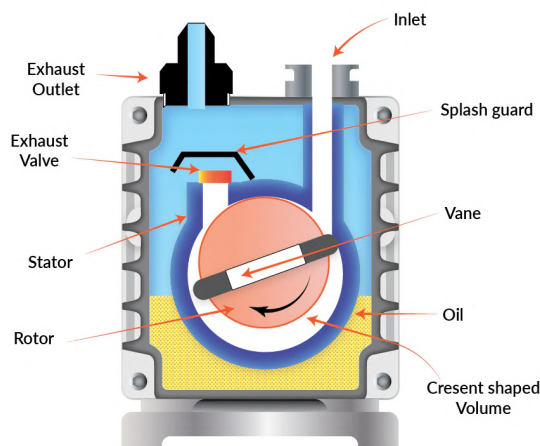


Figure 14: Rotary or Backing Pump

A **rotary vane pump** is a type of positive displacement pump widely used for creating low to medium vacuum levels, typically in the range from atmospheric pressure down to about 10^{-3} mbar. It is commonly used as a *backing pump* for high-vacuum systems like turbomolecular or diffusion pumps, which require a pre-evacuated environment to operate efficiently.

The basic construction of a rotary vane pump consists of a cylindrical housing (called a *stator*), an eccentrically mounted rotor, and a set of sliding vanes that move within radial slots in the rotor. The rotor is offset from the center of the stator, so as it rotates, the distance between the rotor and the stator wall varies. The vanes are pushed outward by centrifugal force or springs, and they maintain contact with the inner wall of the stator, creating sealed compartments (or chambers) between each pair of vanes.

The working principle is based on *increasing and decreasing volumes* within these compartments. As the rotor turns, the gas enters the expanding chamber through the intake port at the inlet side. As rotation continues, the gas is trapped between two vanes and carried around the pump chamber. On the opposite side of the pump, the chamber volume decreases, compressing the gas and forcing it out through the exhaust valve. This continuous process results in a steady suction at the inlet and a compressed gas output at the outlet.

To enhance sealing and reduce wear, many rotary vane pumps use **oil** as a sealing, lubricating, and cooling medium. The oil fills the microscopic gaps between the vanes and the chamber walls, thereby improving vacuum performance and reducing mechanical friction. Some designs use internal oil circulation systems, and the oil also helps capture and condense contaminants and vapors from the pumped gas.

Oil-sealed rotary vane pumps are the most common type due to their ability to achieve deeper vacuums (as low as 10^{-3} mbar) and higher durability under harsh conditions. The oil serves several purposes: it lubricates moving parts, fills micro-clearances

for better sealing, helps to dissipate heat, and traps certain chemical vapors. However, these pumps require regular maintenance, such as oil replacement and filtration, and may suffer from *oil backstreaming* (where oil vapors travel back into the vacuum system if proper traps aren't used). They are generally used where a clean, dry vacuum is not critical.

Dry (oil-free) rotary vane pumps, on the other hand, do not use oil in the pumping chamber. Instead, they rely on self-lubricating vanes made of materials like graphite, and tighter manufacturing tolerances to ensure a seal. While they are easier to maintain and eliminate the risk of oil contamination, they generally do not achieve as deep a vacuum as oil-sealed pumps and may have shorter lifetimes due to wear of dry vanes. Dry pumps are preferred in applications where contamination from oil is unacceptable—such as in cleanrooms, analytical instrumentation, or semiconductor fabrication.

Oil-sealed rotary vane pumps offer higher vacuum levels and better longevity for general-purpose applications, while dry rotary vane pumps offer cleaner operation with easier maintenance but with limitations in ultimate pressure and service life.

13 Turbomolecular Pump

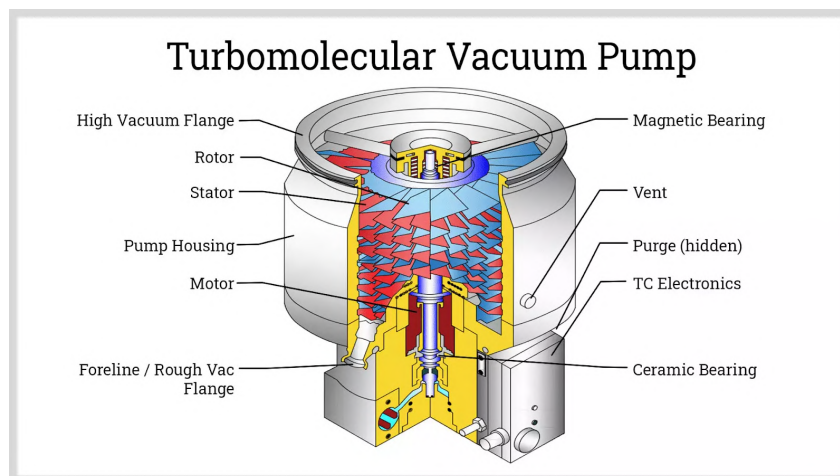


Figure 15: Turbomolecular Pump

A turbomolecular pump is a high-vacuum pump used to evacuate gases from a chamber to pressures as low as 10^{-10} mbar or lower. It is a mechanical, momentum-transfer pump that operates based on the principle of molecular momentum transfer, where rapidly spinning rotor blades impart directional momentum to gas molecules, effectively pushing them from the inlet (low-pressure side) to the exhaust (high-pressure side). Turbomolecular pumps are particularly effective in the high-vacuum regime, where gas flow transitions from viscous flow to molecular flow—that is, where the mean free path of gas molecules becomes larger than the dimensions of the chamber or pump itself.

The core structure of a turbomolecular pump consists of multiple stages of rotor and stator blades arranged in alternating layers. The rotor blades are mounted on a central shaft and are rotated at extremely high speeds, typically ranging from 20,000 to 90,000 revolutions per minute (rpm), depending on the pump design. These blades are inclined at specific angles to the direction of gas flow, optimized for maximum momentum transfer. The stator blades are stationary and are positioned between rotor blades to redirect the gas molecules towards the next rotor stage in a coherent direction, facilitating further compression and transport.

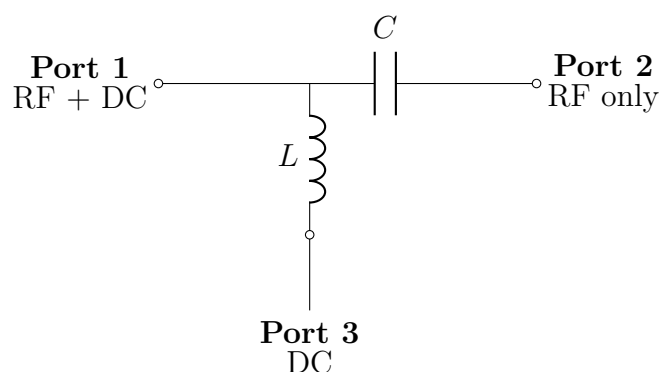
The fundamental operating principle relies on the difference in velocity between the spinning rotor blades and the thermally moving gas molecules. When a gas molecule enters the pump through the inlet, it collides with the fast-moving rotor blades. Because the rotor blades move tangentially at high velocities, they impart a forward momentum to the gas molecule, pushing it in the direction of the pump axis—towards the exhaust. Each collision with the rotor contributes a small directional push. The stator blades then guide the molecules into the correct trajectory to encounter the next set of rotor blades. This sequence of collisions and redirections continues through multiple stages, gradually compressing the gas molecules and moving them down to the exhaust end, where they are typically pumped away by a backing pump, such as a rotary vane or scroll pump, which handles the lower vacuum ranges (typically from atmospheric pressure down to 10^{-2} or 10^{-3} mbar).

Turbomolecular pumps are highly effective at removing light gases like hydrogen and helium, which are difficult to evacuate using other types of pumps. However, their efficiency drops sharply at higher pressures due to increased molecular collisions (viscous flow regime), which interfere with the momentum transfer mechanism. This is why a turbomolecular pump always requires a backing pump to first bring the pressure down to the high-vacuum region before the turbopump can operate efficiently.

Another important aspect of turbomolecular pump design is the precision required in balancing and bearing the rotor. Due to the extremely high rotation speeds, the rotor must be manufactured with high accuracy and dynamically balanced to prevent vibration. Many modern turbopumps use magnetic or ceramic bearings to reduce friction, wear, and contamination, enabling cleaner and more reliable operation, especially in ultra-high vacuum (UHV) environments.

In terms of operation, a turbomolecular pump typically starts with the chamber already pre-pumped by a roughing pump. Once the pressure falls below a certain threshold (usually below 1 mbar), the turbopump is switched on and begins spinning up to its operating speed. The pumping speed and compression ratio vary depending on the design, gas species, and number of rotor-stator stages. Control electronics regulate the motor speed, monitor bearing temperature, and ensure safe startup and shutdown.

14 Bias T



A Bias-T (or Bias Tee) is a passive three-port network used to inject or extract DC power into or from an RF signal path without disrupting the RF signal. This technique is essential in systems where both DC and RF signals must be transmitted over the same coaxial cable. It eliminates the need for separate power lines and enables remote powering of devices. Bias-Ts are widely used in applications such as low-noise amplifiers (LNAs), active antennas, laser diode drivers, and RF-over-fiber systems.

Working Principle of a Bias-T

The working principle of a Bias-T relies on the frequency-selective behavior of inductors and capacitors. Capacitors are reactive components that block DC but allow high-frequency (RF) signals to pass with minimal impedance. Inductors, on the other hand, allow DC to pass while presenting high impedance to RF signals. This difference in behavior is exploited to combine or separate RF and DC signals without mutual interference.

A Bias-T consists of three ports: the RF port, which carries only the RF signal; the DC port, which handles only DC power; and the RF+DC port, which is the combined output or input carrying both signals. In the combining or injection mode, DC power is applied to the DC port and RF to the RF port. At the RF+DC port, the signals merge. The inductor in the DC path prevents the RF signal from entering the DC power source, while the capacitor in the RF path blocks the DC component from reaching the RF source. Consequently, the combined signal appears cleanly at the RF+DC port.

In the separating or extraction mode, a combined RF+DC signal enters the RF+DC port. The inductor routes the DC component to the DC port while blocking the RF signal. Meanwhile, the capacitor routes the RF signal to the RF port and blocks the DC. This separation is made possible due to the inductor's high impedance to RF and low impedance to DC, and the capacitor's inverse behavior.

A basic Bias-T circuit includes an inductor in series with the DC path and a capacitor in series with the RF path. The RF and DC signals meet at a node connected to the RF+DC port. The inductor L allows DC to pass but blocks RF, acting as a high impedance at RF frequencies. The capacitor C allows RF to pass but blocks DC, acting as an open circuit at low frequencies.

When designing a Bias-T, several key parameters must be considered. The first is the

frequency range. The capacitor must have sufficiently low reactance at the lowest operating RF frequency, and the inductor must provide high impedance at the highest RF frequency. Mathematically, the conditions to be satisfied are:

$$X_C = \frac{1}{2\pi f C} \ll 50 \, \Omega \quad \text{and} \quad X_L = 2\pi f L \gg 50 \, \Omega$$

Additionally, the inductor and the circuit traces must be capable of handling the required DC current without overheating or magnetic saturation. Capacitors must have voltage ratings higher than the maximum expected DC bias voltage.

Another important design consideration is insertion loss, which should be kept as low as possible across the RF band to ensure signal integrity. Impedance matching is also crucial; the Bias-T must maintain a characteristic impedance of 50 ohms throughout to avoid signal reflections. Finally, good isolation between the RF and DC paths is necessary to prevent cross-interference between the two signals.

Construction (Making) of a Bias-T

Constructing a Bias-T involves selecting appropriate high-frequency components and assembling them on a suitable platform. Materials typically include high-Q capacitors such as ceramic or mica types and inductors with air or ferrite cores optimized for RF use. Coaxial connectors such as SMA or BNC are used for signal interfacing. The Bias-T may be built on a PCB with controlled impedance traces or constructed using point-to-point soldering for simplicity.

The construction steps begin with PCB design, where microstrip or stripline traces are designed for 50-ohm impedance. The capacitor is placed in-line with the RF path, and the inductor branches from the DC port to the RF+DC junction. Ground planes are included to minimize parasitic effects and ensure impedance control.

Next, the capacitor is soldered between the RF port and the RF+DC port, and the inductor is connected from the DC port to the RF+DC junction. Care is taken to ensure proper grounding and trace layout. After assembly, the circuit is tested using a vector network analyzer to measure insertion loss and VSWR over the desired frequency range. A multimeter or oscilloscope can be used to confirm correct DC routing and signal isolation.

For stability and interference protection, the Bias-T is usually housed in a metallic enclosure. This shield is grounded to system ground to suppress EMI and maintain clean signal performance.

As a practical example, consider a Bias-T operating from 10 MHz to 3 GHz. A suitable design might include a 100 pF high-Q ceramic capacitor and an inductor with a value ranging from 470 nH to 1 μ H. These values provide a reasonable compromise between bandwidth and isolation while ensuring proper functionality across the target frequency range.

15 Attenuator



Figure 16: Attenuator

Passive attenuators are two-port resistive networks designed to reduce signal power from a source to a level suitable for the connected load. They do this without requiring any external power supply, as they are purely passive devices. Attenuators are commonly used in radio, communication, and transmission systems to weaken strong signals, extend measurement range, match impedances, or isolate circuit stages.

They can be fixed, variable, or switchable in steps, with fixed types often called “attenuator pads,” available in standard values from 0 dB to over 100 dB. Variable and stepped attenuators use calibrated resistor networks to allow controlled signal reduction, often in set increments like -2 dB or -6 dB per step.

An attenuator ideally reduces signal strength uniformly across all frequencies without introducing distortion or phase shift. To achieve this, they are built with non-inductive resistors, avoiding reactive components like wirewound resistors that could affect frequency response.

Attenuators are essentially the opposite of amplifiers—they reduce the strength, or gain, of a signal rather than increasing it. The most basic form of a passive attenuator is a resistive voltage divider, where the output voltage is a fraction of the input voltage. The level of attenuation is defined by the ratio of output to input. For example, if a circuit receives 1 volt at the input and delivers only 1 millivolt at the output, the attenuation ratio is:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{1 \text{ mV}}{1 \text{ V}} = 0.001$$

This means the signal has been reduced by a factor of 1,000. However, working directly with such small numbers can be cumbersome, especially in electronic and communication systems. That’s why we commonly express attenuation using a logarithmic scale, specifically in decibels (dB). This simplifies comparisons and makes small and large ratios easier to handle. The decibel is a dimensionless unit (meaning it has no physical unit like volts or watts) and represents the logarithmic ratio of two values—usually voltage, current, or

power. For voltage-based measurements, attenuation in decibels is calculated using the formula:

$$\text{Attenuation (dB)} = 20 \cdot \log_{10} \left(\frac{V_{\text{out}}}{V_{\text{in}}} \right)$$

So, if the output voltage is half of the input voltage:

$$20 \cdot \log_{10}(0.5) = -6 \text{ dB}$$

This negative sign indicates a reduction in signal level. Some standard values to remember:

- A voltage ratio of 0.707 corresponds to -3 dB (approximately 50% power loss).
- A voltage ratio of 0.5 corresponds to -6 dB .
- A voltage ratio of 0.1 corresponds to -20 dB .

In practical attenuator design, the input signal is typically treated as the reference point, often assigned a value of 0 dB. Any drop from that input is reflected as a negative dB value. This reference system helps simplify how we express and understand attenuation. Let's go through an example where a passive attenuator has an insertion loss of -32 dB , and the output voltage is 50 mV. We are asked to find the input voltage.

Using the formula:

$$\text{Attenuation (dB)} = 20 \cdot \log_{10} \left(\frac{V_{\text{out}}}{V_{\text{in}}} \right)$$

Substitute the known values:

$$-32 = 20 \cdot \log_{10} \left(\frac{50 \text{ mV}}{V_{\text{in}}} \right)$$

Divide both sides by 20:

$$-1.6 = \log_{10} \left(\frac{50 \text{ mV}}{V_{\text{in}}} \right)$$

Now take the antilogarithm (base-10):

$$\frac{50 \text{ mV}}{V_{\text{in}}} = 10^{-1.6} \approx 0.0251$$

Solving for V_{in} :

$$V_{\text{in}} = \frac{50 \text{ mV}}{0.0251} \approx 1.99 \text{ V} \approx 2.0 \text{ V}$$

So, to get a 50 mV output with -32 dB of attenuation, the input voltage must be approximately 2.0 V. This use of decibels makes it easier to manage signal levels in electronics, especially when working with signals that span a wide range of values. It's also helpful in communication systems, where dB values can be quickly added or subtracted to account for losses and gains throughout a signal path.

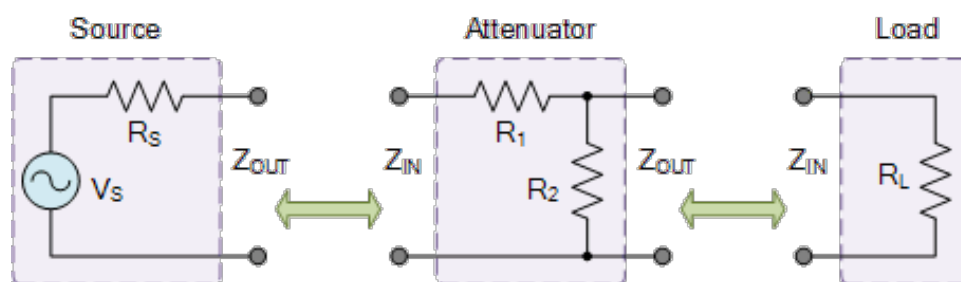


Figure 17: Attenuator Connection

The L-type attenuator is the simplest form of attenuator and consists of two resistors arranged in an “L” shape. It is used primarily for **impedance matching and fixed attenuation** when the source and load impedances are unequal.

It includes:

- A **series resistor** connected in line with the signal path.
- A **shunt resistor** connected from the output node to ground.

This configuration is compact and efficient for low attenuation levels but is not symmetrical, hence only suitable for unidirectional use.

T-type Attenuator

The T-type attenuator consists of three resistors arranged in the shape of the letter “T”. It is used when the input and output impedances are **equal and known**.

- Two **series resistors** are placed on the input and output paths.
- A **shunt resistor** is connected between the junction of the series resistors and ground.

This design allows for better control of signal attenuation and impedance matching, and it is well-suited for **medium to high attenuation levels**.

π -type Attenuator

The π -type attenuator also uses three resistors but in a configuration resembling the Greek letter π . It is effective for impedance matching when a low source resistance is connected to a high load resistance, or vice versa.

- Two **shunt resistors** connect the input and output terminals to ground.
- A single **series resistor** connects the input and output directly.

This structure is particularly advantageous at higher frequencies due to better RF behavior and lower insertion loss.

Balanced vs. Unbalanced Attenuators

Unbalanced Attenuators

In unbalanced attenuators, resistive elements are connected only on one side of the transmission line, with the other side being grounded. This grounded side, called the **common line**, carries no resistive elements and helps in preventing signal leakage at higher frequencies. Unbalanced networks are simple, cost-effective, and commonly used in coaxial systems where one conductor is grounded.

Balanced Attenuators

Balanced attenuators use an equal number of resistors connected symmetrically on both sides of the transmission line. The ground is placed at the center point between the mirrored halves, formed by parallel resistive branches. Balanced attenuators are preferred in differential signal applications and offer **improved noise immunity**.

Special Balanced Versions

- The **balanced version of the T-type** attenuator is known as the **H-pad**.
- The **balanced version of the π -type** attenuator is called the **O-pad**.
- **Bridged T-type attenuators** are also available, combining the T configuration with a bridging resistor to enhance frequency response or isolation.

It is important to note that **balanced and unbalanced attenuator types are generally not interchangeable**. Connecting them together can lead to grounding issues, where one half of the balanced network may be shorted through the unbalanced circuit, causing improper operation or signal distortion. Each type of attenuator—L, T, and π —serves a specific function based on application requirements such as impedance matching, signal level control, and frequency response. Additionally, selecting between balanced and unbalanced configurations is crucial depending on whether the system is single-ended or differential. Proper design and matching ensure minimal signal distortion and optimal system performance.

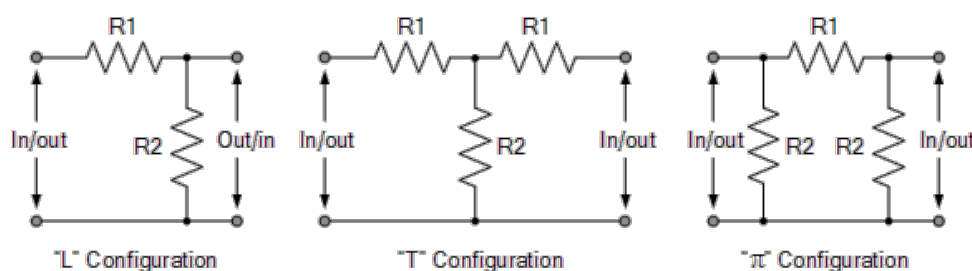


Figure 18: Types of Attenuators

16 Connectors

1. SMA (SubMiniature Version A) Connectors



Figure 19: SMA Connector

SMA connectors are arguably the most widely used RF connectors in cryogenic physics and quantum computing labs. These are coaxial, threaded RF connectors with a 50-ohm impedance. The male connector has a center pin and a threaded outer conductor that screws into the female connector. This threading ensures mechanical reliability, which is critical under thermal cycling conditions in cryostats.

SMA connectors are typically constructed from stainless steel or beryllium copper with gold plating, both of which tolerate cryogenic contraction. They retain mechanical integrity and electrical performance from room temperature down to millikelvin temperatures. These connectors are often paired with semi-rigid coaxial cables such as stainless steel or NbTi to reduce thermal conduction and maintain RF shielding.

Their frequency range spans from DC to approximately 18 GHz in standard versions and up to 26.5 GHz with precision models. SMA connectors are widely used for qubit control, readout resonators, and amplifiers (e.g., HEMTs, JPAs) and are commonly found at 4K, 1K, and millikelvin stages of a dilution refrigerator.

They offer excellent impedance matching, a compact profile, widespread availability, and vacuum compatibility, making them highly suitable for cryogenic RF applications.

1. Male SMA: Has external threads and a center pin (male).
2. Female SMA: Has internal threads and a socket (female) for the pin.
3. "Reverse Polarity" refers to a switch in the gender of the inner conductor, not the threads. RP-SMA Male: Has external threads like a standard male, but contains a socket instead of a pin.
4. RP-SMA Female: Has internal threads, but has a center pin (instead of a socket).

2. BNC (Bayonet Neill–Concelman) Connectors



Figure 20: BNC Connector

BNC connectors are commonly used in laboratory environments for general signal connections but are less suited to high-frequency or precision cryogenic applications. These connectors use a bayonet-style coupling mechanism (a twist-lock), making them quick to connect and disconnect. However, they lack the rigidity and frequency stability of SMA connectors.

They can operate at cryogenic temperatures but may suffer from mechanical expansion mismatch and tend to loosen under repeated thermal cycling unless strain-relieved. Their frequency range extends up to 4GHz, though they are typically used below 1 GHz for reliable performance.

BNC connectors are commonly used for temperature sensors, resistive heaters, DC voltage lines, and low-frequency signal wiring. They are easy to handle and effective for low-frequency analog signals, but are bulky and unsuitable for compact or high-frequency systems.

3. SMP (SubMiniature Push-on) Connectors

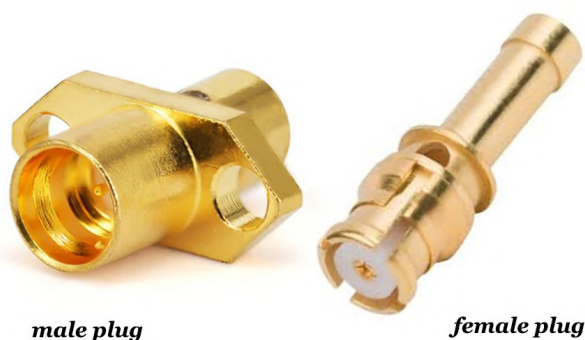


Figure 21: SMP Connector

SMP connectors are snap-in push-on RF connectors designed for high-frequency and high-density systems. They are smaller than SMA connectors and support frequencies up to 40 GHz. The push-on design makes them suitable for board-to-board or module-to-module

applications.

In cryogenic systems, particularly quantum computing cryostats, SMP connectors are often used to accommodate large numbers of qubit lines. These are made from cryo-compatible materials such as gold-plated beryllium copper.

They are ideal for dense RF wiring in multi-qubit systems, amplifier chains, or chip packaging. Their advantages include high signal density, excellent frequency performance, and reduced mechanical stress due to their push-on design. However, they are more expensive, more fragile under thermal cycling, and can be difficult to manage at low temperatures.

4. MMCX / MCX (Micro Coaxial Connectors)



Figure 22: MCX Connector

MMCX and MCX are miniature coaxial connectors used in systems where space and weight are constrained. They are smaller than SMA or BNC connectors and offer a snap-on connection suitable for moderate-frequency performance up to around 6GHz.

In cryogenic setups, these connectors are primarily used in prototypes, portable systems, or test configurations. While they are very compact and lightweight, their mechanical robustness is limited under thermal cycling. They are generally not vacuum-tight and offer limited RF shielding, restricting their use in production systems.

5. N-Type Connectors



Figure 23: N-Type Connector

N-type connectors are larger, threaded coaxial connectors known for good power handling and RF shielding. They are mostly used in room-temperature systems or first-stage cryostat interfaces. While they support frequencies up to 11 GHz, their bulk and high thermal conductivity make them unsuitable for use inside cryostats.

They may be used in external connections, such as signal routing before attenuation or filtering. Their main strengths are ruggedness and high RF performance, but their size and thermal load are major drawbacks for low-temperature applications.

6. D-Sub Connectors (D-Subminiature)



Figure 24: Sub D Connector

D-sub connectors are widely used for DC wiring and signal multiplexing in cryogenic systems. They come in many configurations (e.g., DB-9, DB-15, DB-25, DB-37) with a shielded metallic frame and screw-lock mechanisms.

These connectors are commonly used for thermometry, gate voltages, and control electronics. Only properly rated versions with gold-plated pins and vacuum-tight sealing are suitable for cryogenic use.

D-sub connectors offer high pin counts in a compact format and are quite versatile. However, they provide poor RF shielding and are not suitable for high-frequency signals. They also need to be specifically designed for cryogenic reliability to avoid failure under thermal cycling.

7. Micro-D Connectors

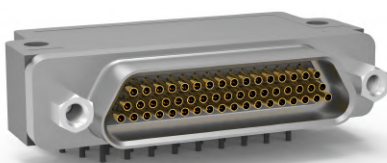


Figure 25: Micro-D Connector

Micro-D connectors are miniaturized versions of D-sub connectors, used when space is at a premium. They are widely used in dilution refrigerators, especially for dense gate wiring in quantum dot and spin qubit experiments.

These connectors are extremely compact and lightweight, leading to lower thermal loads and higher wiring density. On the downside, they require special crimping or soldering tools and are more expensive than standard connectors.

17 Circulator

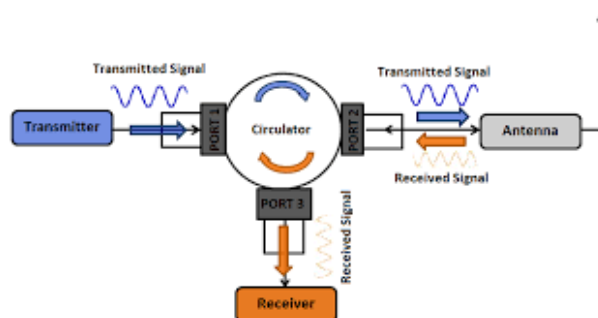


Figure 26: Circulator

A circulator is a specialized passive, non-reciprocal device commonly used in high-frequency and microwave circuits to control the direction of signal flow between multiple ports. Typically consisting of three or four ports, a circulator routes an incoming signal at one port to the next port in a sequential order, either clockwise or counterclockwise, while preventing the signal from flowing back to the previous port. For example, in a three-port circulator, a signal entering port 1 is directed only to port 2, a signal at port 2 is directed to port 3, and a signal at port 3 is directed to port 1. This controlled signal routing enables engineers to isolate different parts of a system from one another without using active components.

The core functionality of a circulator relies on non-reciprocity, meaning the signal path is not the same in both directions. This behavior is achieved using ferrite materials and a magnetic bias field. When high-frequency signals pass through the ferrite under the influence of the magnetic field, they experience a phenomenon called Faraday rotation. This causes the wave energy to rotate and couple preferentially into the next port in the rotation direction, rather than reversing or splitting between multiple paths. The physical design of a circulator, including the placement of transmission lines or waveguides, the geometry of the ferrite disks, and the applied magnetic field, determines the operational frequency range and direction of circulation.

Circulators are particularly important in systems where signal separation is essential but using multiple antennas or complex switching would be impractical. A common application is in radar systems, where a single antenna is used for both transmission and reception. Here, a circulator allows the transmitted signal to pass from the transmitter (connected to port 1) to the antenna (connected to port 2), while ensuring that the received signal from the antenna is directed only to the receiver (connected to port 3), avoiding any backflow into the transmitter. Similarly, in RF testing setups, circulators can protect sensitive signal generators from reflected signals due to impedance mismatches by routing such reflections to a termination load.

18 Keithley



Figure 27: Keithley

For detailed information, one can refer to the *Series 2400 SourceMeter User's Manual* ([click to download](#)).

19 Yokogawa 7651 DC Source



Figure 28: Yokogawa 7651 DC Source

For detailed information, one can refer to the *Yokogawa 7651 Programmable DC Source Instruction Manual* ([click to download](#)).

20 Stanford Research Systems Model SR830 DSP Lock-In Amplifier



Figure 29: Stanford Research Systems Model SR830 DSP Lock-In Amplifier(click to download)

For detailed information, one can refer to the *User's Manual*(click to download).

21 Arbitrary-Waveform-Generator



Figure 30: AWG70000

For detailed information, one can refer to the *AWG70000 series arbitrary waveform generator User Manual* (click to download).